



Explicitly AI-Supported Model Development

Institution

Halle Institute for Economic Research (IWH) and GIZ

Abstract

This document outlines a comprehensive advanced online training program for DGE-METRIC users. The training emphasizes explicitly AI-supported model development using modern tools like VS Code and collaborative workflows. The program spans 4–5 days of interactive sessions, totaling 10–15 hours of instruction and hands-on practice.

Training Overview

Institution: Halle Institute for Economic Research and GIZ

Target Audience: Advanced users with basic experience in Dynare and macroeconomic modeling.

Format: Online, interactive sessions (suggested: 4–5 days, 2–3 hours per session)

Appetizer: VS Code and AI-Assisted Development

Why This Matters

In modern macroeconomic research, efficiency and reproducibility are paramount. This training introduces you to powerful tools that accelerate your workflow and enhance your research productivity.

Key Tools

- **Visual Studio Code (VS Code):** A lightweight, extensible code editor that streamlines your Dynare/MATLAB work with syntax highlighting, debugging, integrated terminals, and version control integration.
- **AI-Assisted Development:** Leverage AI tools (e.g., GitHub Copilot, Claude, and similar assistants) to:
 - Generate and refine model equations and code snippets
 - Automate repetitive coding tasks
 - Debug errors and optimize performance
 - Document your model logic and methodology
 - Accelerate research workflows and reduce development time

Your Benefits

- Faster model development and iteration cycles
- Better code organization, documentation, and maintainability
- Collaborative workflows with version control (Git)
- AI-enhanced productivity and research impact
- Skills applicable across macroeconomic modeling projects

5-Day Training Agenda

Day 1: Deep Dive into DGE-METRIC Structure

1. Recap of DGE-METRIC core blocks and equations
2. Model calibration: advanced parameterization and data sources
3. Customizing the model for country- or sector-specific analysis

Learning Outcomes: Participants will understand the complete structure of the DGE-METRIC model, master calibration techniques, and be able to adapt the model to country-specific contexts.

Day 2: Advanced Scenario Design

1. Building complex policy scenarios (multi-policy, shocks, time paths)
2. Using Excel input files: structure, best practices, and troubleshooting
3. Hands-on: Creating and loading custom scenarios

Learning Outcomes: Participants will design realistic policy scenarios, work efficiently with input data management, and troubleshoot common scenario development issues.

Day 3: Model Extensions and Sensitivity Analysis

1. Adding new sectors, technologies, or climate modules
2. Sensitivity analysis: parameter sweeps and uncertainty quantification
3. Automating batch runs and result collection

Learning Outcomes: Participants will extend the model with custom features, conduct rigorous sensitivity analysis, and automate analysis workflows for efficiency.

Day 4: Advanced Output Analysis

1. Interpreting simulation results: macro, sectoral, and distributional impacts
2. Visualization: advanced plotting in MATLAB/Excel
3. Exporting results for policy briefs and reports

Learning Outcomes: Participants will extract meaningful insights from model results, create professional visualizations, and communicate findings effectively to policymakers.

Day 5: Open Lab and Q&A

1. Participants work on their own cases with instructor support
2. Troubleshooting, tips, and sharing best practices
3. Wrap-up and feedback

Learning Outcomes: Participants will receive personalized guidance on their research applications and solidify their understanding through practical problem-solving.

Prerequisites

- Working Dynare and MATLAB installation on your computer
- Basic knowledge of DGE-METRIC structure (e.g., from the introductory training or documentation)

- Familiarity with macroeconomic modeling concepts (stocks, flows, equilibrium conditions)
- Recommended: Basic Git and VS Code experience (or willingness to learn these during the training)

Training Agenda Summary

The table below provides a quick reference for all five training days:

Day	Topics and Learning Outcomes
Day 1	Structure: Recap core blocks, advanced calibration, model customization
Day 2	Scenarios: Complex policy scenarios, Excel data management, hands-on practice
Day 3	Extensions: Add sectors/technologies, sensitivity analysis, automation
Day 4	Analysis: Result interpretation, advanced visualization, report generation
Day 5	Lab: Participant projects, troubleshooting, best practices, feedback